VAROLLINE INTERNSHIP PROGRAM | AI ENGINEER TASK

Smart Text Classifier Dataset Documentation

Labelled short-text dataset for Complaint, Inquiry, Feedback and Other classification

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1. Dataset Overview

This dataset contains **1,200 short English text messages**, each labelled with one of four categories. It is used to train and evaluate a Logistic Regression classifier (TF-IDF features) that predicts the category of a customer-style message. The file is stored as `data/text_classification_dataset.csv` with three columns.

Column	Type	Description
id	Integer	Unique row number (1 to N).
text	String	The short message written by a user.
category	String	Target label: Complaint, Inquiry, Feedback or Other.

2. Data Source and Creation Method

No public dataset was copied. The dataset was **created specifically for this task** with a reproducible Python script (`src/generate_dataset.py`, random seed 42). The process was:

1. Hand-written samples: 12 realistic messages per category, including borderline sentences (for example an unhappy message that is really a question).

2. Template generation: many additional messages produced from sentence templates combined with varied slot values such as products, cities, issues, features, days and times.

3. Noise injection: about 15% of generated samples were lower-cased or had their final punctuation removed, to imitate messy real-world input.

4. Cleaning and balancing: duplicates were removed, each class was balanced to the same size, and rows were shuffled.

3. Category Definitions

Category	Definition	Example
Complaint	The user reports a problem or expresses dissatisfaction and expects a fix, refund or action.	The product stopped working after two days. This is a waste of money.
Inquiry	The user asks a question or requests information (price, timing, steps, availability).	Do you offer a student discount on the annual plan?
Feedback	The user shares an opinion, praise or a suggestion without demanding urgent action.	It would be nice to have a dark mode option in the app.
Other	Greetings, announcements, acknowledgements, spam and anything outside the three classes above.	Good morning everyone.

4. Dataset Statistics

Category	Samples	Share	Avg. words	Min words	Max words
Complaint	300	25%	17.1	6	26
Inquiry	300	25%	10.0	5	16
Feedback	300	25%	12.0	7	20
Other	300	25%	8.8	2	15
Total	**1200**	**100%**	**12.0**	**2**	**26**

The classes are perfectly balanced, so accuracy is a fair metric alongside macro F1-score. There are **0** missing values and **0** duplicate texts.

5. Sample Rows

ID	Text	Category
3	The app keeps crashing whenever I log in. Please escalate this to a manager.	Complaint
5	Once again I have an issue: my account was suspended without any notice. I will cancel my service if this continues.	Complaint
12	I want to report a serious problem: the internet connection drops every few minutes. I expect a full refund.	Complaint
13	I am not happy at all because the internet connection drops every few minutes. I expect a full refund.	Complaint
2	I would love to see a mobile version of the website in the next release	Feedback
10	Nice work on the product quality, the team should be proud.	Feedback
11	Nice work on the pricing structure, the team should be proud.	Feedback
14	The website design is great, and the professional team made it even better.	Feedback
4	Can I pay for the laptop in instalments	Inquiry

ID	Text	Category
7	How do I return a headphones?	Inquiry
9	Can you let me know when the new version will be released?	Inquiry
17	Can you let me know whether the smart watch comes with a warranty?	Inquiry
1	The team meeting is scheduled for Tuesday at 3:30 pm	Other
6	Your one-time code is 968735. Do not share it with anyone.	Other
8	Congratulations! You have won a brand new phone. Click the link to claim it now.	Other
41	reminder: the training session starts on thursday at 3:30 pm.	Other

6. Labelling Guidelines

• A message asking for something to be fixed, refunded or escalated is a **Complaint**, even if polite.

• A message whose main purpose is to ask for information is an **Inquiry**, even if the tone is negative.

• Praise, opinions and improvement ideas that do not demand urgent action are **Feedback**.

• Greetings, announcements, acknowledgements, spam and random text are **Other**.

7. Limitations of the Dataset

• Most samples are template-based, so vocabulary and sentence patterns are less diverse than real user text.

• Only English is covered; Roman Urdu or mixed-language messages are not included.

• Each message has exactly one label; real messages can contain several intents at once.

• High scores on this dataset may be optimistic; the model should be re-tested on genuine user messages.